Research Gaps and Enhancements — EnHi-LMP Project

Oct 13, 2026 · @Manish

CSET 346 – Natural Language Processing · Explainable Multilingual Legal Meaning Preservation Using JUDGEBERT

Summary

The project addresses five gaps in the JUDGEBERT base paper: one language, one domain, a small dataset, no out-of-domain test and no explanation of scores. Its main result: legal-aware scoring reaches Pearson r 0.60 against the rubric-based LMP labels, while the best generic similarity metric reaches 0.37.

The base paper is Beauchemin et al., *JUDGEBERT: Assessing Legal Meaning Preservation Between Sentences* (EMNLP 2025). It scores whether a simplified legal sentence keeps the original's legal meaning on a 1–10 scale, using 297 French insurance pairs (FrJUDGE). This project extends that work to English and Hindi, adds tenancy law, and makes every score explainable.

Research gaps in the base paper

Nine gaps were found: five are named as limitations of JUDGEBERT in the project brief, and four more were identified while building this project.

#	Research gap	Why it matters	Where it comes from
G1	Only one language (French)	No evidence the approach works for other languages, especially Indian languages written in Devanagari	Named in the brief
G2	Only one legal domain (property-damage insurance)	Legal wording differs by domain; a metric may not carry over to contracts such as rent agreements	Named in the brief
G3	Small dataset (297 pairs)	Too little data to train or test a robust model	Named in the brief

#	Research gap	Why it matters	Where it comes from
G4	Sentences judged without surrounding context	A clause can depend on definitions or exceptions elsewhere in the document	Named in the brief
G5	No out-of-domain evaluation	Unknown whether the model generalises beyond its training data	Named in the brief
G6	Score only, no explanation	A user cannot see *what* changed (deadline, party, exception) or trust the score	Identified in this project
G7	No typed legal-error categories	Errors such as negation, modality or number changes are not labelled, so they cannot be measured separately	Identified in this project
G8	No test of generic similarity on Indian-language legal text	English-centric similarity tools may fail on Hindi without anyone noticing	Identified in this project
G9	No public English–Hindi legal meaning-preservation dataset	Indian-language legal simplification cannot be evaluated at all	Identified in this project

Enhancements made, gap by gap

Seven gaps are fully addressed, one partly (G3, until human scoring is done), and sentence-level context (G4) is left for future work.

Gap	Enhancement in this project	Where to find it	Status
G1 One language	Parallel English + Hindi data; all models and features work in both languages	Slides 5–6, 8; paper Sec. IV	Done
G2 One domain	Added tenancy / rent-agreement law (40 clauses) beside insurance (40 clauses)	Slide 6; paper Sec. IV	Done

Gap	Enhancement in this project	Where to find it	Status
G3 Small data	320 new pairs + 320 sanity pairs; two-annotator workbooks and agreement script ready	Slide 6; code `annotation/`	Partly: human scoring still to do
G4 No context	Not addressed; planned paragraph-level input	Paper Sec. IX–X	Open
G5 No out-of-domain test	Cross-lingual (EN↔HI), cross-domain (insurance↔tenancy) and zero-shot test on FrJUDGE human labels	Slide 9; paper Sec. VII-B/C	Done
G6 Score only	Every pair gets a score, a 3-way decision and a one-line rationale	Slides 6, 10; paper Sec. V-B	Done
G7 No error types	10-type legal error taxonomy, each pair tagged; error-type precision/recall/F1 reported	Slides 3, 10; paper Table VI	Done
G8 No Hindi test of generic similarity	Showed embedding similarity drops from r 0.38 (English) to 0.21 (Hindi) and scores unrelated Hindi pairs 6.6/10	Slides 8–9; paper Sec. VII	Done
G9 No EN–HI legal dataset	Built EnHi-LMP and released it with the code	Code `data/`	Done

Additional technical enhancements

Beyond the gaps, eight design choices make the system stronger, fairer to evaluate and easier to reproduce.

Enhancement	What it does	Why it matters
LegalDiff features	19 bilingual legal-difference signals: modality, negation, numbers, durations, parties, conditions, exceptions, scope	Gives the model explicit legal signal; strongest single component (r 0.60)

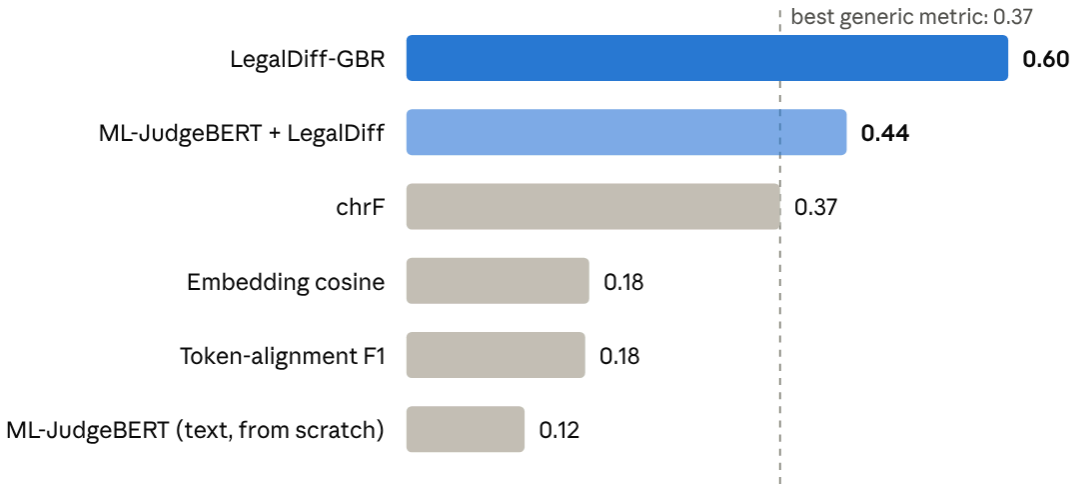
Enhancement	What it does	Why it matters
Rule-based explainer	Turns LegalDiff signals into an error type and a rationale, e.g. “Amount changed: 5000 → 50000”	Makes each score auditable
ML-JudgeBERT + LegalDiff fusion	JUDGEBERT-style cross-encoder with legal features fused before the regression head	Lifts the transformer from r 0.12 to 0.44
Indian number handling	Reads Devanagari digits, “15,00,000” and “15 lakh” as the same amount	Prevents false “number changed” errors in Indian legal text
Grouped cross-validation	English and Hindi twins of a clause always stay in the same fold	Stops translation leakage from inflating results
Calibrated baselines	Each similarity score is mapped to the 1–10 scale on training folds only	Makes RMSE comparable between baselines and models
Held-out sanity sets + ablation	320 identical/unrelated pairs never used in training; model also run without augmentation	Checks the scale ends honestly, as JUDGEBERT recommends
Reproducibility kit	Colab notebook for XLM-R / MuRIL, annotation workbooks, Krippendorff’s α script, Git-ready code	Lets the remaining steps be completed and verified

Evidence that the enhancements work

LegalDiff-GBR reaches r 0.60, beating chrF (0.37), the best generic metric; the transformer trained from scratch only becomes useful once LegalDiff is fused in.

Legal-aware scoring beats every generic similarity metric

Pearson r with LMP labels, 5-fold grouped CV, 320 EN+HI pairs (mean of 3 seeds)



Experiment 1 results (results/exp1_summary.json) · 5-fold grouped CV, n = 320

Enhancement tested	Result	Gap closed
Hindi support	LegalDiff stable: r 0.60 English, 0.59 Hindi; embedding cosine drops 0.38 → 0.21	G1, G8
Cross-lingual transfer	Trained on English, tested on Hindi: r 0.48; reverse: 0.54	G5
New domain	Insurance → tenancy r 0.55; tenancy → insurance r 0.57	G2, G5
Real human labels (FrJUDGE, French)	Trained only on EnHi data: r 0.56–0.59; embedding cosine −0.17	G5
Sanity checks	Identical pairs 9.99/10, unrelated 1.22/10; embedding cosine gives unrelated Hindi pairs 6.6/10	G3, G8
Explanations	Error detected with F1 0.79; exact error type named for 40% of errors	G6, G7

All EnHi numbers use rubric-based silver labels; the FrJUDGE row is the only one scored by independent human annotators.

Gaps still open and future work

Six gaps remain; the first two are already prepared and are the priority before final submission.

Remaining gap	What is needed	Ready-made support
Labels are rubric-based, not human	Two annotators score all 320 pairs; report Krippendorff's α and re-run experiments	Annotation workbooks A/B + `merge_annotations.py`
Transformer trained from scratch (r 0.12 alone)	Fine-tune XLM-R and MuRIL on a GPU	Colab notebook `ML_JudgeBERT_pretrained_Colab.ipynb`
No document context (G4)	Feed the surrounding clause or definitions with each sentence	Planned
Weak typing of scope changes and added content (F1 0.00 and 0.15)	Better scope and hallucination detection, e.g. from a pretrained encoder	Planned
Negation hidden in a verb change ("does not cover" → "pays for") missed	Semantic polarity detection, not only negation words	Planned
Each corrupted pair has exactly one controlled error	Test on real, LLM-generated simplifications with several errors	Planned